Application Management with GitOps

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Self introduction

Name: Wanja Pernath

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Base: Germany (very close to the Alps)

Role: EMEA Technical Partner Development Manager - OpenShift and MW

Experience: Years of Consulting, Training, PreSales at Red Hat and before

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LinkedIn:https://www.linkedin.com/in/wanjapernath/

GitHub: https://github.com/wpernath

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First book just published

Getting GitOps

A technical blueprint for developing with Kubernetes and OpenShift based on a REST microservice example written with Quarkus

Technologies discussed:

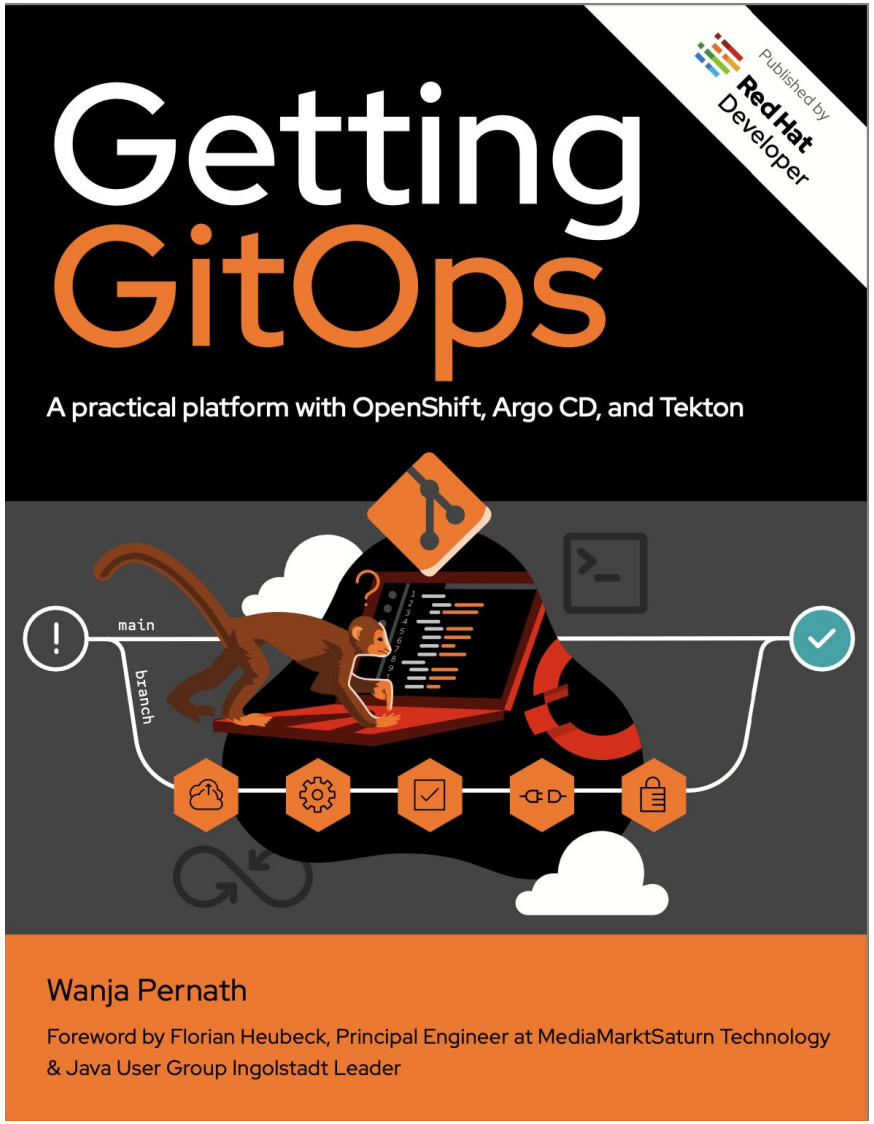
Quarkus, Helm Charts, Kustomize, Tekton Pipelines, Kubernetes Operators, OpenShift Templates, ArgoCD, CI/CD, GitOps....

Download for free at:

https://developers.redhat.com/e-books/getting-gitops-practical-platform-openshift-argo-cd-and-tekton

Interview with full GitOps Demo:

https://www.youtube.com/watch?v=znMfVqAIRzY&ab_channel=OpenShift



Agenda

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Agenda

- Demo Overview
- Outlook on next sessions

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Demo Overview

Demo Sources:
https://github.com/wpernath/quarkus-grumpycat

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Challenge

- Applications are getting more and more complex
- Deployment of a simple application can't be fully automated
- This results in poor software quality (deployment)
- Deployments are not repeatable

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Pain points

- For this we are using different CI and CD tools
- Sometimes they are not flexible enough to be used in our processes
- Others are over engineered or too complex

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The Solution

- OpenShift Pipelines is a fully integrated and flexible layered product on top of Red Hat OpenShift to do Continuous Integration (CI)
- OpenShift GitOps is a fully integrated and flexible layered product on top of Red Hat OpenShift to do Continuous Deployments (CD)
- Red Hat Quay is an enterprise container registry and repository

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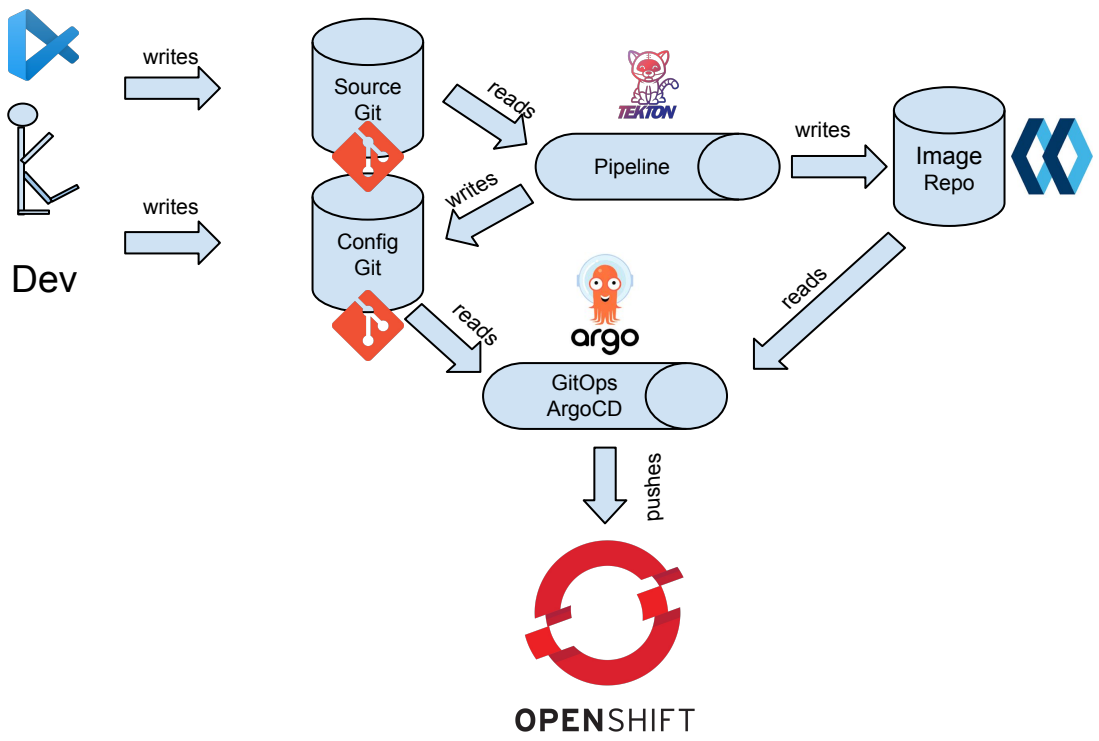
Demo



- Setting up a typical development environment with just a few steps
- Writing code changes
- Start CI-Pipeline
- CI-Pipeline writes into Config Git Repository and into Image repository
- Argo CD reads from Config Git Repository and from Image Repository

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Episode #1

Installation of the Application via GitOps

Episode

#1

- One simple CLI call to the OpenShift cluster fully installs the application
- It will be installed as DEV, STAGE and PROD
- Each with a different configuration
- Everything is centrally managed in a Git environment

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Episode #2

Build and Dev Workflow

Episode

#2

- Every push to Git will start the build pipeline
- This pipeline builds the sources and the container image
- It will also update the configuration Git repository

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Episode #3

Preparing Staging Workflow

Episode

#3

- On request the current Dev environment will be branched in Git
- It's becoming the Staging environment
- The container image will be tagged with same tag as config in Git
- The pipeline will create a GitHub pull request
- Deployment happens as soon as the pull request is merged into main branch (manual approval)

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Episode #4

Preparing Production Workflow

Episode

#4

- On request the current Staging environment will be branched in Git
- It's becoming the Production environment
- The container image will be tagged with same tag as config in Git
- The pipeline will create a GitHub pull request
- Deployment happens as soon as the pull request is merged into production branch (manual approval)

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Benefits

- Quick deployments
- Repeatable deployments
- Deployment quality is better
- Everything is automated

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Outlook

- Using GrumpyCat to deeper understand Quarkus and the benefits of it
  - Using Extensions
  - Creating REST services
  - Using Dev Services
- Understanding base technologies behind GitOps
  - Kustomize
  - Helm Charts
  - Tekton / OpenShift Pipelines
  - Argo CD

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Thank you

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